Rational and irrational number along with its kinds and types along with examples some p/q numbers are also irrational how

Let's understand rational and irrational numbers in a very simple and clear way.

🔵 I. Rational Numbers

✅ Definition:

A number is rational if it can be written in the form:

$$\frac{p}{q}$$

Where:

- p and q are integers
- $q \neq 0$

✅ Important Property:

Their decimal form is either:

- Terminating (ends)
- Repeating (recurring pattern)

🔹 Types of Rational Numbers

1️⃣ Integers

All integers are rational because:

$$5 = \frac{5}{1}$$

Examples:

- -3
- 0
- 7

2️⃣ Fractions (Proper & Improper)

- Proper: $\frac{3}{5}$
- Improper: $\frac{7}{4}$

3️⃣ Terminating Decimals

Examples:

- 0.5 = 1/2
- 0.25 = 1/4
- 0.125 = 1/8

4️⃣ Repeating (Recurring) Decimals

Examples:

- 0.3333... = 1/3
- 0.6666... = 2/3
- 0.142857142857... = 1/7

🔴 2. Irrational Numbers

✅ Definition:

A number is irrational if:

- It cannot be written as p/q
- Its decimal is non-terminating and non-repeating

🔹 Types of Irrational Numbers

1️⃣ Square Roots (of non-perfect squares)

Examples:

- √2
- √3
- √5

Because they never end and never repeat.

2️⃣ Cube Roots (of non-perfect cubes)

Examples:

- ∛2
- ∛7

3️⃣ Famous Mathematical Constants

- pi = 3.1415926535...
- Euler's number = 2.718281828...

They never repeat or terminate.

🔵 Important Question:

❓ Can some p/q numbers be irrational?

👉 No. Never.

If a number can be written exactly as:

$$\frac{p}{q}$$

(where p and q are integers)

Then it is ALWAYS rational.

❗ Then why confusion happens?

Because:

Example:

$$\frac{\sqrt{2}}{2}$$

It looks like a fraction (something over something).

But...

√2 is irrational.

So:

$$\frac{\sqrt{2}}{2}$$

is still irrational.

Because:

- Rational ÷ Irrational = Irrational
- Irrational × Rational = Irrational

🔴 Important Rule Summary

Form	Type
p/q (p,q integers)	Rational
Terminating decimal	Rational
Repeating decimal	Rational
Non-terminating non-repeating decimal	Irrational
√ of non-perfect square	Irrational
π, e	Irrational

🔵 Quick Examples Test

Number	Rational / Irrational
5	Rational
3/7	Rational
0.75	Rational
0.121212...	Rational
√9	Rational (because 3)
√2	Irrational
π	Irrational

If you want, I can also explain:

- Why √2 is irrational (proof)
- Why repeating decimals are rational (proof)
- Or number system hierarchy diagram (N, W, Z, Q, R)

Just tell me 😊